

Misspecification Aliases a Calibrated Error Bar: A Bi-exponential IVIM Fit Loses *Conditional* Coverage of the Trusted Diffusion Coefficient Inside the Identifiable Region

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Abstract

A deployed quantitative-MRI parameter map carries an error bar. Split-conformal prediction makes that bar *marginally* valid for free, even when the fitted model is wrong [1, 2]. We show, on a fully synthetic intravoxel-incoherent-motion (IVIM) digital reference object, that this guarantee hides a failure that matters in practice: fitting a *bi-exponential* model on *non-bi-exponential* truth leaves marginal coverage of the well-identified tissue-diffusion coefficient D intact (0.909 at nominal 0.90), yet breaks its *conditional* coverage along a latent, heavy-tailed perfusion-misspecification axis (conditional gap +0.105, 95% CI [+0.093, +0.117]). The failure is mechanistically distinct from a within-model identifiability limit: under the *exact* partition a companion identifiability study uses to define its “trust D ” region [3]^P — the bottom-two terciles of true D^* — the gap does not vanish but *intensifies* (well-identified gap +0.148, [+0.131, +0.164]; strictly inside the identifiable region, +0.196), while in that study’s own ill-posed high- D^* regime the gap is near zero (+0.014). The driver is a signed bias of $\hat{D}$ that grows $\sim 7.7\times$ with departure as the heavy-tailed perfusion term aliases into the high- b tissue slope. We pair the characterization with an observable, ground-truth-free diagnostic that reads the bi-exponential fit’s residual *structure*; it detects the heavy-tail channel (AUC 0.684, [0.673, 0.694]), beating a naive deployment-drift monitor (0.550), but—reported honestly—pure perfusion dispersion stays a near-hidden channel (AUC 0.578, below the pre-registered 0.60 floor): the contribution is a *heavy-tail misspecification detector*, not a universal one. The separation is robust across SNR 35–100, cohort sizes, and b -schemes (11/12 conditions), with one stated boundary: it disappears at SNR 25, where noise dominates the bias. All results are seed-generated with no clinical data and make no clinical claim.

1 Introduction

Quantitative MRI increasingly ships a parameter *map* with a per-voxel uncertainty. Conformal prediction is an attractive way to put a distribution-free guarantee on that uncertainty: under exchangeability, split-conformal intervals attain a user-specified *marginal* coverage in finite samples for *any* underlying predictor, including a biased or misspecified one [1, 2]. A practitioner could

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reasonably read this as a safety net: even if the IVIM bi-exponential model is not exactly the physics, the conformal error bar still covers at the nominal rate.

That reading is a trap. Marginal coverage is an average over the cohort; it can be held exactly while coverage *conditional* on a clinically relevant axis fails badly [4]. The question this paper answers is concrete: when the fitted model is *wrong*—bi-exponential fit, non-bi-exponential truth—which conditional axis does the conformal bar fail along, does that failure touch the parameter practitioners are told to *trust*, and can a deployer see it coming from observables alone?

We make three contributions, each pre-registered with a refute condition and reported with a bootstrap confidence interval.

1. **The misspecification instance (§4.1).** On a synthetic IVIM digital reference object, a heavy-tailed (stretched-exponential) perfusion departure leaves the *marginal* coverage of the well-identified tissue coefficient D at nominal but breaks its *conditional* coverage along the latent departure axis. The mechanism is a signed bias of $\hat{D}$ that grows $\sim 7.7\times$ as the heavy perfusion tail aliases into the high- b tissue slope; a faster tri-exponential pool, which decays *away* from high b , produces no such failure (a pre-registered null).
2. **Distinct from identifiability (§4.2).** The failure is *not* a restatement of the within-model identifiability wall that afflicts D^* at high D^* [3]^P. Under that study’s own partition of the latent axis, the gap on D survives and *intensifies* inside the identifiable region it calls safe, while it is near zero in the ill-posed regime—an *opposite* gradient. Misspecification ($\neq$ identifiability) of a *different*, well-identified parameter is the wedge.
3. **An observable diagnostic, honestly scoped (§4.4).** A label-free reading of the bi-exponential fit’s residual *structure* detects the heavy-tail channel (beating a naive deployment-drift monitor), while pure dispersion stays near-hidden. We lock the claim to a *heavy-tail misspecification detector*, not a universal one.

The throughline is that model misspecification is invisible from inside the model: the marginal guarantee holds, the trusted parameter’s point estimate looks ordinary, and only a structured residual—or knowledge of the latent departure—reveals that the error bar has been silently aliased.

2 Background and related work

Conformal marginal robustness (the baseline we confirm, not the contribution). Split-conformal prediction yields finite-sample marginal coverage under exchangeability for any base predictor [1, 2]; the guarantee is robust to model misspecification *marginally*. Our marginal result (§4.1) is a confirmation of this classical fact and is explicitly *not* the contribution; we report it to delimit where the guarantee stops.

Conditional coverage is not free. Barber et al. [4] prove that exact distribution-free *conditional* coverage is impossible without further assumptions. This is the theoretical reason a latent axis *can* carry a coverage failure invisibly; our work is a concrete, mechanistic IVIM instance of that impossibility under model misspecification.

Identifiability of D^* vs. misspecification of D . A companion line of work [3]^P establishes that the IVIM pseudo-diffusion coefficient D^* is conditionally under-covered in its *high- D^** tercile because the Fisher information for D^* collapses there (a Cramér–Rao identifiability wall): the model is assumed *correct*, and the failure is a property of estimating D^* . Our failure is orthogonal

on both axes: it is the coverage of the *tissue* coefficient D (the parameter that study marks well-identified and trustworthy), it occurs *because the model is wrong*, and it lives in the *low- D^** region that study calls safe. §4.2 makes the distinction quantitative under that study’s exact partition.

Misspecification diagnostics in qMRI. Goodness-of-fit and model-order selection (AIC/BIC/ F -test for mono/bi/tri/stretched IVIM) ask which model to fit; a misspecified-Cramér–Rao variance-consistency test has been proposed for arterial spin labeling [5]. Our diagnostic differs in target: it does not pick a model order, it predicts *where the conditional coverage of a trusted parameter fails*, from the deployed bi-exponential fit’s residual structure, in IVIM/diffusion rather than ASL. IVIM uncertainty quantification by calibration curves and CRPS [6] addresses a different question (aleatoric/epistemic UQ) with no conformal coverage and no misspecification axis.

3 Methods

Synthetic ground truth (Lattice DRO). All cohorts are drawn from a clean-room, seed-generated IVIM digital reference object [7]^P; no measured data are used and no clinical claim is made. The canonical signal is the Le Bihan bi-exponential [8]

$$S(b)/S_0 = f e^{-bD^*} + (1 - f) e^{-bD},$$

with base parameters (D, D^*, f) drawn from a documented physiological prior and *seed-locked across the departure sweep*, so the same voxel population is observed under increasing departure. Three non-bi-exponential perfusion families replace the single D^* exponential while leaving the tissue term $(1 - f)e^{-bD}$ —hence D —an exact ground-truth parameter: *stretched-exponential* ($e^{-(bD^*)^\beta}$, heavy tail; departure β), *log-normal dispersion* (a gamma/log-normal mixture; departure CV), and *tri-exponential* (a faster third pool; departure g). Each reduces *exactly* to the bi-exponential at its limit ($\beta = 1$, CV = 0, $g = 0$), a built-in placebo.

Estimator and intervals. Every voxel is fit by a bounded bi-exponential nonlinear least squares—deliberately misspecified whenever the truth is not bi-exponential. We form split-conformal absolute-residual intervals for D at $\alpha = 0.10$ (nominal 0.90). A single *departure-blind* radius, calibrated on the pooled placebo-and-worst-departure cohort, defines both the marginal coverage and the conditional gaps, so the marginal claim and the gap claim share one calibration recipe.

The well-identified- D^* subset = the companion’s identifiable region. To test distinctness rigorously, we define the “well-identified” subset by the *exact* construction the companion identifiability study uses for its latent axis: terciles of true D^* via `np.quantile([1/3, 2/3])` followed by `np.digitize`. The identifiable region (where that study says “trust D ”) is the bottom-two terciles; the top tercile is its ill-posed wall. We report the gap on the full test set, on the bottom-two terciles ($n \approx 333$), on the strict bottom tercile ($n \approx 164$, the deepest interior of the identifiable region), and—for contrast—on the top tercile.

Observable diagnostic and baseline. From the bi-exponential fit residual (no ground truth) we read three structure statistics—reduced- χ^2 , lag-1 autocorrelation, and longest same-sign run—and score a worst-vs-placebo detection AUC. The baseline-to-beat is a naive label-free deployment-drift monitor (a Mahalanobis score on observable signal-shape summaries plus the fit-residual norm, the standard recipe), *computed on the same cohorts* rather than asserted.

Table 1: Conditional gap of D under the companion study’s *exact* latent-axis partition (true- D^* terciles). The gap survives and intensifies inside the identifiable region (“trust D ”) and is near zero in the companion’s own high- D^* wall—an opposite gradient. Heavy-tail family; brackets are 95% cluster-bootstrap CIs (16 seeds).

D^* region	role for the companion study	conditional gap of D
all voxels	—	+0.105 [+0.093, +0.117]
bottom-2 terciles (well-ID)	identifiable; “trust D ”	+0.148 [+0.131, +0.164]
strict bottom tercile	deepest in identifiable region	+0.196 [+0.170, +0.220]
top tercile (high D^*)	the companion’s ill-posed wall	+0.014 [+0.001, +0.027]

Statistics. Headline gaps carry two-level (seed, voxel) cluster-bootstrap 95% CIs over 16 seeds; AUCs and the bias ratio carry across-seed 95% CIs over 8 seeds. Every number in this manuscript is regenerated into `numbers.tex` by `paper/consistency.py` from the seeded gate outputs.

4 Results

4.1 Marginal coverage holds; conditional coverage of D breaks

Under the departure-blind radius, marginal coverage of D stays at nominal across all three families (0.909, CI [0.900, 0.919] for the heavy-tail family; nominal 0.90): the conformal guarantee is intact on average, confirming [1]. Yet for the heavy-tail (stretched) family the *conditional* coverage of D degrades monotonically along the latent departure axis, for an overall conditional gap of +0.105 (CI [+0.093, +0.117], 16 seeds). The mechanism is a signed bias of $\hat{D}$ that grows $\sim 7.7\times$ from placebo to worst departure: the heavy perfusion tail persists to high b and aliases into the tissue slope, biasing $\hat{D}$ in a direction the departure-blind interval cannot anticipate. The bi-exponential-limit placebo row itself stays at nominal, so the failure is departure-driven, not a binning artifact (refute R3 cleared).

4.2 Distinct from a within-model identifiability wall

The central claim is that this is not the companion study’s D^* identifiability wall in disguise. Table 1 reports the gap on D under that study’s *own* partition of the latent D^* axis. Inside the identifiable region (bottom-two terciles, where the companion says “trust D ”) the gap does not collapse toward the marginal—the pre-registered refute R2—but *grows* to +0.148 (CI [+0.131, +0.164]), and grows further, to +0.196, in the strict bottom tercile deepest inside that region. In the companion’s *own* ill-posed high- D^* regime, the gap is near zero (+0.014). The gradient is therefore *opposite* to an identifiability wall (Fig. 1): identifiability fails where D^* is large; our coverage failure of D concentrates where D^* is small (slow perfusion decay leaks furthest into the high- b tissue slope). Misspecification of a well-identified parameter, not unidentifiability, is the operative cause.

4.3 The wedge is mechanism-specific, not generic misspecification

Not all misspecification breaks D -coverage. Table 2 contrasts the three families. The faster tri-exponential pool decays *away* from the high- b tissue slope and produces no conditional failure (+0.000 overall, -0.009 well-identified—a pre-registered null). Pure log-normal dispersion leaks only mildly into high b and gives a weak break (+0.040). The pattern tracks the high- b aliasing

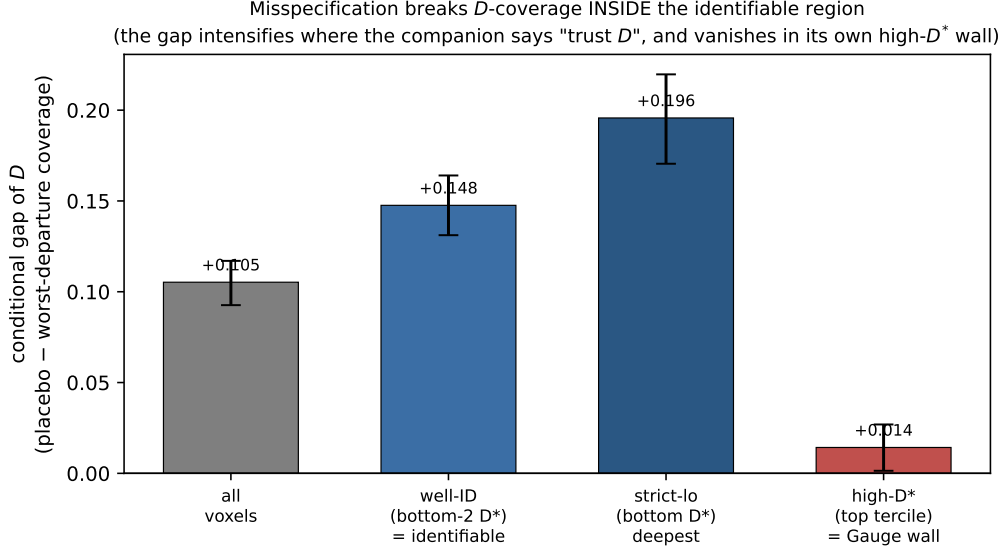


Figure 1: Conditional gap of the trusted tissue coefficient D by true- D^* regime. The gap is largest deep inside the companion study’s identifiable region (“trust D ”, low D^*) and near zero in that study’s own ill-posed high- D^* wall—the *opposite* of an identifiability gradient. Bars are bootstrap point estimates; whiskers are 95% CIs.

Table 2: Mechanism-specificity across departure families. Only the heavy-tail channel breaks the conditional coverage of D ; the faster tri-exp pool is a null, dispersion is weak. (Conditional gap of D ; 8 seeds.)

family	pre-registered	overall gap	well-ID gap
stretched-exp (heavy tail)	break	+0.105	+0.148
log-normal dispersion	weak	+0.040	+0.052
tri-exponential (faster pool)	null	+0.000	−0.009

mechanism precisely: only departures that push perfusion signal into the tissue-slope regime bias $\hat{D}$ and break its conditional coverage.

4.4 An observable diagnostic, honestly scoped

Can a deployer detect the misspecification—and thus the risk to D ’s conditional coverage—without ground truth? Reading the bi-exponential fit’s residual *structure* detects the heavy-tail channel at AUC 0.684 (CI [0.673, 0.694]; the structure statistics alone, lag-1 autocorrelation and longest-run, give 0.682), above the residual-magnitude channel alone (0.670) and well above the naive deployment-drift monitor (0.550, CI [0.543, 0.556]), and it ranks voxels by their realized $|D\text{-error}|$ (Spearman $\rho = 0.169$). Reported honestly, the reach is bounded: pure dispersion stays a near-hidden channel (AUC 0.578, below the pre-registered 0.60 detection floor, consistent with the naive monitor also failing there), and the tri-exp pool is *detectable* (AUC 0.627) yet harmless to D —detection and coverage-failure coincide only for the heavy-tail channel. We therefore lock the claim to a **heavy-tail misspecification detector**, not a universal one.

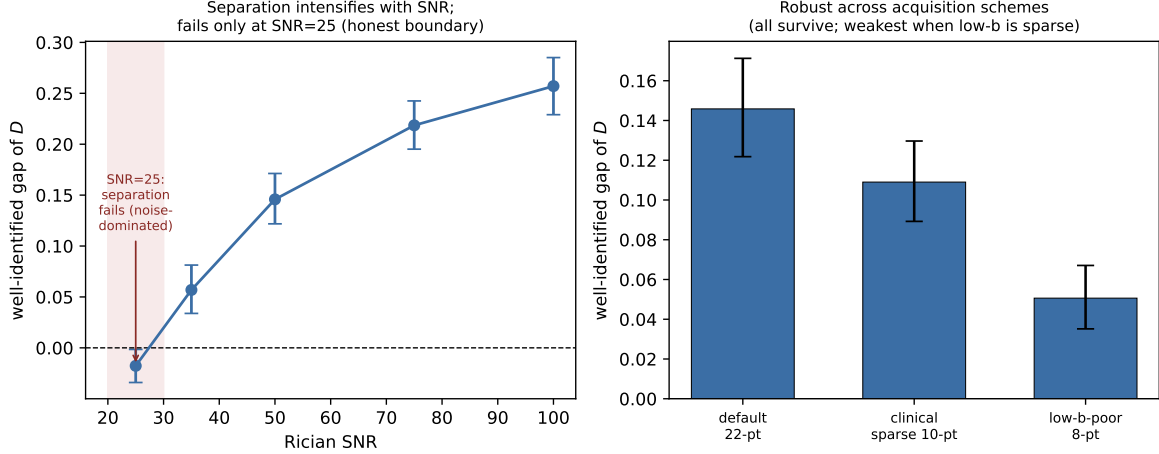


Figure 2: Robustness of the heavy-tail separation. *Left*: the well-identified gap of D grows with SNR and survives down to SNR 35, but vanishes at SNR 25 (shaded; the honest boundary). *Right*: the gap survives across acquisition schemes, weakening when low- b sampling is sparse (consistent with the perfusion-aliasing mechanism). Whiskers are 95% CIs.

4.5 Robustness envelope and an honest boundary

The heavy-tail separation holds across the realistic envelope: it survives in 11 of 12 swept conditions—SNR 35–100 (the well-identified gap *grows* with SNR, to +0.257 at SNR 100), cohort sizes 1000 up to 2000 (+0.154 at $n=2000$), and acquisition schemes from the dense default to a sparse clinical 10-point scheme (+0.109). The one boundary, reported rather than buried: at SNR 25 the separation disappears, because there the Rician-floor noise dominates the departure-driven bias (Fig. 2). The effect is thus a property of the moderate-to-high-SNR regime (default SNR 50) where IVIM diffusion mapping is actually trusted.

5 Discussion

A distribution-free *marginal* guarantee can coexist with a large, clinically-shaped *conditional* failure of the very parameter a deployer is told to trust—when the fitted model is wrong. We exhibit this as a sharp, mechanistic IVIM instance: heavy-tailed perfusion misspecification aliases into the high- b tissue slope, biases the well-identified D , and breaks its conditional coverage *inside* the region a within-model identifiability analysis marks safe. The effect is the mirror image of an identifiability wall—same modality, different parameter, opposite D^* gradient, and a different root cause (misspecification, not lost Fisher information).

What this is not. It is not a claim that conformal prediction “breaks”: marginal coverage is held throughout, exactly as the theory promises [1]. It is not the D^* identifiability wall [3]: §4.2 separates them under that study’s own definition. It is not a model-order selector or an ASL variance test [5]; it predicts where a trusted parameter’s conditional coverage fails. And it is not a universal misspecification detector: the diagnostic is scoped to the heavy-tail channel.

Limitations. The study is fully synthetic by design (a digital reference object isolates the mechanism without confounds); it makes *no clinical claim*, and the diagnostic is moderate (AUC 0.684)

and family-dependent. Pushing the diagnostic (structured-residual generalized likelihood ratios, multi-segment consistency) and an in-vivo demonstration on an open cohort are future work. The honest SNR boundary (§4.5) bounds the regime of validity.

Reproducibility

Every number is seed-generated and regenerated by `bash reproduce.sh`, which re-runs the three gate drivers (`experiments/run_phase{1,2,3}.py`), writes `results/phase{1,2,3}_*.json`, and then `paper/build.sh` regenerates `numbers.tex` via `paper/consistency.py` (which also checks that every manuscript macro is defined and the locked-scope invariants hold) and compiles this PDF. Ground truth is the Lattice digital reference object [7]^P (seed-generated; no data files). No private or clinical data are touched.

Forward-citation note. Gauge [3] and Lattice [7] are in-program references and are unpublished at the time of writing; no published DOI is asserted for either. Each is swapped to its published reference at finalization per the checklist in `release_config.json`.

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